



Isaac Vargas

Software Engineer

✉ hello@isaacvargas.me

🌐 https://isaacvargas.dev

📞 +1 (951) 716-2297

📍 California, US

SUMMARY

I am a chemist turned software engineer. I've written software that keeps scientific, manufacturing, and operations workflows moving smoothly. Seeing how code can solve very specific puzzles is what keeps me motivated to build; There are always new challenges and opportunities to learn and apply that knowledge.

EXPERIENCES

Software Engineer

May 2022 - Present

Essentials by Catalina, Inc.

California

- Built and maintain Lumexia, a manufacturing ERP unifying procurement, inventory, production, QC, pricing, and R&D — now running production for two cosmetic manufacturing facilities. Next.js, TypeScript, PostgreSQL, Prisma, RustFS, Docker.
[View Lumexia Website](#)
- Developed parametric pricing tooling that recalculates costs against material availability and commodity swings, preventing over \$400,000 in mispriced production runs.
- Designed validation workflows that catch formulation and batch errors before production, reducing corrective actions by 92%.
- Automated compliance and formulation reporting for chemists and regulators, cutting manual data prep from 52 hours to 2.
- Own deployment and CI/CD across facilities via Docker and GitHub Actions.

Formulations Chemist, then Software Developer

Jan 2017–Feb 2022

Maki Creations, Inc.

California

- Built internal utilities automating lab calculations and parsing supplier technical documents, replacing manual R&D workflows for a team of three.
- Formulated cosmetic products and established QC protocols, applying analytical chemistry standards to testing methods.

PROJECTS

<https://isaacvargas.dev>

UNCI to UNII | Next.js, Typescript, Prisma, Tailwind, RadixUI, Docker

2024 - Current

A Cosmetic Ingredient Compliance Tool that helps prepare FDA compliant formulation data.

- Automated INCI-to-UNII resolution for FDA cosmetic registration against the FDA public API, with persistent custom mappings for edge-case ingredients.
- Exports portal-ready CSV matching the Cosmetics Direct import schema, eliminating manual cross-referencing during MoCRA submissions.

Woolrus | Typescript, WooCommerce, Docker, Shipstation API

2025 - Current

A pick, pack and verification software for software that purchases and generates shipping labels.

- Implemented the end-to-end fulfillment workflow — station-based pick/pack/QA steps, PDF shipping-label generation via pdf-lib, S3-compatible object storage for order media, fuzzy SKU search, and full localization with next-intl.
- Integrated ShipStation and automated querying shipping options and saving on average 25 to 35 percent on every package by programmatically choosing the cheapest options.

Helm | Next.js, Typescript, DaisyUI, SQLite, Docker

2026

Helm organizes your links into tabs, sections, and bookmarks, and lets you restyle almost every pixel. You can change gradients, themes, background images, per-element colors and borders, keyboard shortcuts, and more.

EDUCATION

Bachelor of Science, Chemistry | California State University, Fullerton

2016

- While pursuing my BSc Chemistry, I worked as an undergrad researcher in Dr. Nicholas T. Salzameda organic synthesis laboratory. I developed and validated organometallic peptidomimetics synthetic pathways for potential remedial drugs for the Botulinum neurotoxin type A serotype light chain.

SKILLS

Languages: TypeScript, JavaScript, Go, Python, Bash, LaTeX, Lua

Data: PostgreSQL, Redis, SQLite, S3-compatible object storage

Frameworks: React, Next.js, Prisma, Tailwind CSS, Radix UI

Infrastructure: Docker, GitHub Actions, CI/CD, Linux, Cloudflare, Nginx